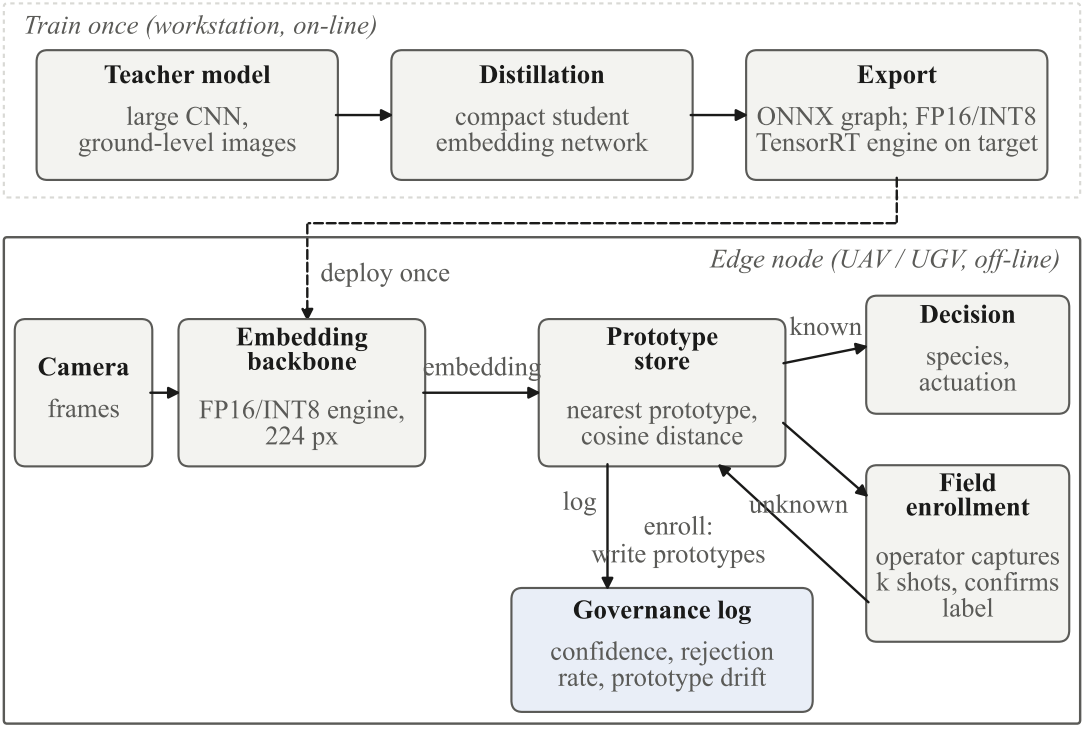
Train once (workstation, on-line)
Teacher model
large CNN, ground-level images
Distillation
compact student embedding network
Export
ONNX graph; FP16/INT8 TensorRT engine on target
deploy once
Edge node (UAV / UGV, off-line)
Camera
frames
Embedding backbone
FP16/INT8 engine, 224 px
embedding
Prototype store
nearest prototype, cosine distance
known
Decision
species, actuation
log
enroll: write prototypes
unknown
Field enrollment
operator captures k shots, confirms label
Governance log
confidence, rejection rate, prototype drift